

Rishab Kumar

[LinkedIn](#), [GitHub](#) [HackerRank](#)

Email: kumarrishab2910@gmail.com

Mobile: +91-7023036378

EDUCATION

- **Indian Institute of Technology, Gandhinagar** Jul 2022 - Jun 2024
Master of Technology - Mechanical Engineering; CPI: 8.06/10
- **National Institute of Technology, Nagaland** Jul 2018 - May 2022
Bachelor of Technology - Mechanical Engineering; CGPA: 9.42/10

EXPERIENCE

- **Robotics Control and Perception Engineer** Flux Auto, Bengaluru(India)
April 2026 - Present
 - **Path Tracking** : Developed lateral and longitudinal path tracking algorithms such as Stanley, Pure Pursuit, and LQR controllers for BOPT and mining vehicle.
 - **Velocity Profile** : Developed the velocity profile for a BOPT to optimize acceleration, velocity ensuring cargo stability and safe braking distances.
 - **Pallet Detection** : Implemented a perception pipeline using LiDAR data and template matching for accurate pallet detection and localization.
- **Robotics Engineer** Alphadroid, Noida(India)
May 2025 - Nov 2025
 - **Sensor Fusion**: Improved EKF-based odometry and integrated it into the stack for better pose estimation.
 - **Visual Odometry**: Developed a visual odometry pipeline for RGB-D cameras and integrated it with an EKF for improved state estimation.
 - **Motion Planning**: Improved the navigation stack by reducing end point accuracy, settling time and by integrating a custom behavior tree node.
 - **Local Planner**: Integrated and tuned local planners, including DWB and MPPI, through extensive testing.
 - **Sensor Data**: Identified and resolved IMU and camera data issues, significantly enhancing localization and navigation performance.
- **Junior Research Fellow** Suman Mashruwala Advanced Micro Engineering Lab, IIT Bombay(India)
July 2024 - Feb 2025
 - **Algorithm**: Developed a precision-focused printing algorithm for a micro 3D printer (*Spanda1*).
 - **Control**: Implemented control algorithms such as PID and LQR for X and Y stage actuation.
 - **Debugging**: Performed circuit debugging and sensor interfacing for the micro 3D printer.
- **Teaching Assistant** IIT Gandhinagar(India)
Aug 2022 - Apr 2024
 - **Introduction to Robotics**: Graded assignments and projects for 25+ students, assisted with hardware setup, and provided expert debugging support on Orangewood cobot.
 - **Manufacturing Laboratory**: Ensured safety protocols and aided students in machining, supported course projects and grading for over 100 students.

PROJECTS

- **M.Tech Thesis** Human-Centered Robotics Lab, IIT Gandhinagar(India)
Jan 2023 - Jun 2024
 - **Hardware Setup**: Design and Developed a 2R Cable-Driven Manipulator hardware setup for upper limb assistance.
 - **Control Algorithm**: Designed a high level position control algorithm to achieve precise end-effector positioning.
 - **Simulation and Hardware Testing**: Simulated in MuJoCo to validate high level control strategies and successfully tested on the hardware.
- **Mobile Manipulator** Advanced Robotics, IIT Gandhinagar(India)
Aug 2023 - Nov 2023
 - Developed a real-life manipulation task as pick up and drop using a mobile manipulator.
 - Integrated a 5-DOF manipulator with a 2-DOF mobile base to enable coordinated mobile manipulation.
 - Utilized the custom made kinematics, dynamics, and control of a mobile manipulator for end-effector.

- **3P Manipulator**
Oct 2022 - Nov 2022

Introduction to Robotics, IIT Gandhinagar(India)

- Designed a 3P-joint manipulator integrating vision-based object detection with precise position control.
- Implemented a vision-guided nut detection and unscrewing using real-time image processing and precise position control.

SKILLS SUMMARY

- **Programming Languages:** C++, Python
- **Robotics Middleware & Frameworks:** ROS1, ROS2, Nav2, MoveIt2
- **Simulation & Control:** MuJoCo, Gazebo, MATLAB, Simulink
- **OS & Tools:** Linux, Docker; Git
- **Sensors:** IMU, LiDAR, RGB-D Camera, Load Cell, Encoder
- **Technical Skills:** Robot Kinematics & Dynamics, Control Systems, Motion Planning, and Perception.

HONORS AND AWARDS

- Awarded the Gold Medal for outstanding student for achieving the first rank in the B.Tech Mechanical Engineering program (2018-2022).
NIT Nagaland(India) – Jan 2023